

Investigate how cultural perspectives and world views influence the development of scientific knowledge

Learning intention

- investigate how cultural perspectives and world views influence the development of scientific knowledge.

Teach and model

- investigating how collaboration between Swami Vivekananda and Nikola Tesla, who held different world views.
- investigating how world views about the role of women lead to women scientists being placed in subordinate roles and 'written out' of history.
- analysing how world views relating to fairness in sport have led to the development of rapid chemical tests to identify performance-enhancing drugs.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.